

LONDON-CORBIN, KY, USA

WMO: 724243

Lat: 37.087N

Lon: 84.077W

Elev: 1188

StdP: 14.08

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 03849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	16.3	0.4	5.9	16.3	5.3	7.5	20.1	21.7	46.9	19.5	44.8	4.3	290	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	90.5	72.8	88.4	72.5	86.5	71.8	76.3	85.3	75.2	83.8	74.2	82.4	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	130.1	80.0	72.5	125.9	79.0	71.7	122.4	78.3	40.4	85.8	39.3	84.1	38.4	82.4	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	15.9	12.9	DB	4.0	93.9	6.5	4.0	-0.7	96.7	-4.5	99.0	-8.2	101.3	-12.9	104.1	
			WB	3.0	78.2	6.3	1.7	-1.5	79.4	-5.2	80.4	-8.8	81.3	-13.4	82.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	36.0	39.5	47.3	57.2	65.4	72.7	75.7	74.9	69.0	57.5	46.5	39.4
	DBStd	16.39	11.76	10.37	10.39	8.84	7.16	4.98	3.84	4.25	6.41	8.38	8.98	10.21
	HDD50	1540	455	319	177	34	3	0	0	0	0	27	169	356
	HDD65	4161	900	713	552	258	87	8	0	1	35	259	555	793
	CDD50	4039	20	26	94	250	479	680	797	773	570	258	65	27
	CDD65	1185	0	0	4	24	98	237	332	309	154	25	2	0
	CDH74	9885	0	1	42	275	857	1947	2665	2500	1332	254	12	0
	CDH80	3220	0	0	4	39	194	631	950	901	453	48	0	0
Wind	WSAvg	4.9	6.1	6.2	6.3	6.1	4.7	4.2	3.8	3.4	3.4	3.9	4.7	5.5
Precipitation	PrecAvg	45.6	3.5	3.3	4.3	4.1	4.3	3.5	5.0	3.3	3.5	2.8	3.9	3.9
	PrecMax	62.0	8.5	5.6	9.2	8.6	11.0	6.5	10.2	7.4	7.5	7.7	8.5	11.6
	PrecMin	33.7	1.1	0.6	1.3	0.6	1.1	0.7	1.1	0.8	0.9	0.0	0.4	0.4
	PrecStd	8.0	2.0	1.3	2.2	1.7	2.5	1.6	2.6	1.6	1.5	1.6	1.8	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.5	70.8	79.3	84.3	87.4	91.2	93.2	94.1	91.5	84.9	75.7	69.4
		MCWB	58.3	60.1	61.5	63.4	69.6	72.4	74.8	73.3	69.0	68.4	62.7	59.9
	2%	DB	63.1	66.0	73.7	80.5	84.4	88.6	90.0	90.3	88.3	80.4	70.2	64.0
		MCWB	56.0	56.1	58.6	61.9	68.0	72.5	74.3	73.0	70.0	65.8	58.7	57.3
	5%	DB	58.9	61.4	69.3	77.3	82.2	86.4	87.8	87.7	85.4	76.3	66.2	60.0
		MCWB	53.5	52.7	56.4	60.6	67.4	71.5	73.8	72.9	69.7	62.9	56.5	54.0
	10%	DB	54.5	56.9	64.7	73.5	79.3	83.7	85.4	85.2	82.0	72.5	62.4	56.4
		MCWB	49.0	49.4	54.1	59.1	66.5	70.4	72.8	72.2	68.2	62.0	54.3	51.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	61.8	63.3	67.0	73.1	76.6	78.2	77.3	75.0	72.0	64.6	62.0
		MCDB	64.7	68.3	74.7	77.9	81.8	86.0	87.3	87.6	83.8	79.8	71.3	66.9
	2%	WB	57.2	58.3	61.0	65.1	71.2	75.0	76.8	75.7	73.3	68.9	61.6	58.7
		MCDB	62.2	64.1	71.1	75.7	80.3	84.0	85.7	84.9	82.4	75.3	68.6	62.9
	5%	WB	54.5	54.4	58.6	63.1	69.5	73.7	75.5	74.7	71.7	66.2	58.6	55.4
		MCDB	58.7	59.5	66.9	72.4	78.6	82.3	83.7	83.4	79.9	72.9	63.9	59.1
	10%	WB	50.1	50.6	55.8	61.3	68.0	72.4	74.5	73.5	70.2	63.8	55.4	51.7
		MCDB	53.9	55.6	62.8	70.5	76.2	80.5	82.3	81.6	77.9	70.2	60.9	55.9
Mean Daily Temperature Range	5% DB	MDBR	17.3	18.4	20.8	22.5	20.9	19.6	18.1	19.2	21.0	22.5	20.7	17.0
		MCDBR	21.4	24.4	26.5	27.7	24.4	22.6	21.3	22.2	25.7	27.0	25.8	21.7
		MCWBR	15.1	16.4	14.5	12.8	10.5	9.0	7.9	8.1	9.7	13.2	15.3	15.2
	5% WB	MCDBR	18.6	20.5	21.8	21.7	20.6	18.8	17.9	18.5	20.3	20.7	19.2	19.1
		MCWBR	14.4	15.5	13.6	11.7	10.1	8.8	7.5	7.6	8.5	11.6	13.6	15.2
Clear-Sky Solar Irradiance	taub	0.305	0.314	0.347	0.389	0.439	0.457	0.484	0.489	0.404	0.356	0.319	0.301	
	taud	2.508	2.476	2.422	2.301	2.183	2.168	2.115	2.087	2.311	2.441	2.513	2.533	
	Ebn at Noon	282	293	292	283	269	263	255	251	268	273	274	276	
	Edh at Noon	25	30	34	41	47	47	49	50	38	30	25	23	
All-Sky Solar Radiation	RadAvg	645	842	1203	1587	1769	1946	1818	1721	1441	1107	789	565	
	RadStd	68	118	107	107	115	155	109	100	91	148	142	87	57

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	+144	
(47)	Regional (22 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+236	+144	

Nomenclature: See separate page